How to increase Worker Efficiency & Save Time

The Inside Scoop on Beth Marie's Old Fashioned Ice Cream

Beth Marie's Order Fulfillment Process Improvement

Beth Marie's on the Square | Denton, TX

1. Executive Summary

This report presents the completed Define, Measure, Analyze, and Improve phases of a Six Sigma DMAIC project focused on addressing order fulfillment inefficiencies at Beth Marie’s Ice Cream. The store consistently experienced long customer waiting times during peak periods which negatively impacted on the overall customer experience and workers. Our initial evaluation revealed that the store operated depending on customer volume, which means a single staff would work during slow periods and more staff work during busy periods.

The primary objective of this project was to analyze the order fulfillment system using a waiting line (queuing) model and identify the main reason for inefficiency. We divided the total service time into three measurable components which are decision time, scoop time, and checkout time. A structured data collection process was implemented during the Measure phase, where we conducted in-person observations to capture per-customer service times across multiple variables, item type, time of day, and staffing levels. We also conducted a customer perception survey to evaluate actual wait times compared to customer expectations and experiences.

Analysis of the collected data revealed that decision time was the primary bottleneck in the system which was responsible for the largest portion of total service time and contributing significantly to waiting time and longer queues. Customers who were unfamiliar with menu options or undecided on flavor selection required more time at the counter, which slowed the overall flow of service. Scoop time was found to be moderately variable and influenced by order complexity, while checkout time remained relatively consistent under normal conditions. However, during peak periods, checkout time increased due to inefficiencies introduced by the busy-period workflow.

A key finding from the analysis was that the handwritten weight slip used during busy periods caused several types of errors that reduced both speed and accuracy. These problems included wrong weights being read, which led to incorrect charges, delays when cashiers had trouble reading handwriting, and occasional lost slips that required orders to be weighed again. These issues did not happen in the single-staff process, where one employee handled the entire order from start to finish. This shows that the problems came from the process design, not from employee effort or workload.

Further analysis using charts and graphs supported these results. A histogram of total service time showed that most orders took a similar amount of time, but a few took much longer because customers took extra time deciding or placing more complex orders. Trend analysis showed that service times stayed mostly steady over time, meaning there was no major change in worker performance, within a specific methods observation window. Instead, differences were caused by individual customer behavior.

Also, even though items like shakes were ordered less often, they took longer to prepare and slowed down service during busy times.

The customer survey gave more information about how people felt about waiting. It showed that customers who did not already know what flavor they wanted were more likely to feel like they waited a long time, even if their actual wait was normal. This supports the idea that customer decision-making has a big effect on how the service is experienced.

There were also some limits in the data. Queue length was not always recorded which made it harder to fully measure arrival rates in the waiting line model. Also, point-of-sale data was not available, so historical sales and error rates could not be checked. Even with these limits, the data was still enough to find the main problems and make strong conclusions.

Based on the results, several improvements were identified. The biggest opportunity is reducing decision time by making the menu easier to see and understand before customers reach the counter. This could include clearer signs or better displays of flavors. Another improvement is creating a clear and standard process for busy times so staff know exactly what to do and when to switch roles.

Beth Marie’s would benefit from developing a stronger digital presence to address seasonal slowdowns. While the brand is well known locally for its community involvement and long-standing history, foot traffic tends to decline outside of peak summer months and major events on the square.

Leveraging social media offers a cost-effective way to maintain visibility and attract a broader customer base year-round. The business is already well positioned to implement this strategy, as many of its employees are local high school and university students who are likely familiar with current social media trends and content creation through their involvement in campus organizations.

By utilizing these internal resources, Beth Marie’s can consistently promote engaging content such as new and returning flavors, in-store events, and community outreach efforts. This approach not only reinforces brand identity but also keeps the business relevant during slower periods.

Additionally, creating a marketing internship position could further strengthen this effort. With several nearby colleges and a steady demand for internship opportunities, the role would likely attract qualified candidates who can bring fresh ideas and have key insights on how to attract the college demographic.

Lastly, we would suggest adding self-service kiosks. These would let customers choose their order on a screen before reaching the counter. This could reduce decision time and help shorten the line during busy periods.

Another important improvement is removing the handwritten weight slip. Replacing it with a digital system like the self-service kiosks or connecting a digital system directly to the register would reduce mistakes, save time and make checkout faster. It is also important to clearly define staff roles during busy periods so there is less confusion and smoother service.

Overall, the project shows that the main causes of slow service are customer decision time and the way the busy-period process is designed. Employee performance was not the major issue, but more so a lack of consistency within the work structure. By making targeted improvements, the store can reduce wait times, improve accuracy, and give customers a better experience.

This project provides clear data to support improvements and shows how important, good process design is in busy service environments. The recommendations made through the DMAIC process can realistically improve both speed and customer satisfaction.

2. Problem Statement and Business Context

2.1 Problem Statement

Beth Marie's is well known for long lines at its square location in Denton, TX. The store currently operates two structurally different order fulfillment processes depending on customer volume, with no standardized transition between them. During slow periods, a single staff member manages the entire order sequence. During busy periods, roles are split between makers and cashiers; however, the handoff relies on handwritten weight slips that introduce legibility errors, incorrect charges, and added checkout time.

The absence of a consistent, defect-resistant process across both operating modes drives excessive customer wait times and creates avoidable billing errors. This project aims to analyze and model the store's waiting line behavior, identify inefficiencies within the current fulfillment process, and develop actionable recommendations to improve operational throughput and deliver a better guest experience.

2.2 Business Context and Impact

Beth Marie experiences two predictable rush windows each day: a post-school rush and a post-dinner rush. These are the highest-revenue periods of the day, making operational efficiency during these windows critical to business performance. Checkout delays during peak periods compound queue length, potentially deterring customers from entering or returning. Billing errors create direct financial consequences — incorrect charges either overcharge customers, eroding trust, or undercharge, creating revenue leakage.

The store also experiences a seasonal slowdown in winter compared to summer. POS data access, which would enable precise identification of peak times and transaction-level volume data, is pending approval from the store owner. The general manager could not grant access directly and has referred the team to the owner. If access is not approved, proxy measures from direct observation will be used.

A secondary issue — the flavor display system — was identified during the general manager interview but falls outside the current project scope. It may be revisited as a future subset of this work if time permits.

2.3 Key Stakeholders

Stakeholder	Role	Project Relevance
General Manager	Operations Lead	Identified operational problem; primary project contact; open to process changes
Store Owner	Decision Maker	Approves technology investment, POS data access, and customer survey permission
Store Staff (Makers)	Process Executor	Execute order preparation; write weight slips during busy periods
Store Staff (Cashiers)	Process Executor	Interpret weight slips and process payment; primary recipient of handoff defects
Customers	End User	Directly affected by wait times, billing accuracy, and overall guest experience

3. The Five Phases of DMAIC, SIPOC Diagram and Documentation

3.1 The Five Phases of DMAIC

Charter Element	Detail
Project Title	Worker Efficiency & Wait Time Reduction — Beth Marie's on the Square, Denton TX
Problem Statement	Beth Marie's experiences long customer queues and extended service times, particularly during peak periods. The busy-period split-role process relies on a handwritten weight slip handoff that introduces billing errors and checkout delays.
Goal Statement	Apply a waiting line model to quantify service time components (decision, scoop, checkout), identify process bottlenecks, and deliver actionable recommendations to improve order fulfillment efficiency and guest experience.
Project Scope	In Scope: Customer ordering process from queue entry to payment completion; busy-period and slow-period workflows; waiting line model development. Out of Scope: Supply chain, inventory decisions, flavor display system (potential future subset).

Business Case	Long queues risk deterring customers and reducing throughput during the store's highest-revenue periods. Standardizing the fulfillment process and identifying bottlenecks through data-driven analysis will directly improve customer experience and operational efficiency.
Project Team	Kiara (Project Manager), Meshack (Data Analyst), Michael (Outreach & Visual Design), Giovani (AI & Systems Support), Reethika (Data Visualization), Annuska (Data Visualization)
Timeline	Define & Measure phases: current submission. Analyze, Improve, Control phases: subsequent submissions.
Success Metrics	Reduction in average total service time during peak periods; identification of primary bottleneck step; customer survey perception aligned with observed wait times; actionable process recommendations delivered.

3.2 SIPOC Diagram

The SIPOC below maps the busy-period order fulfillment process, which is the primary scope of this project. The slow-period single-staff model is noted for comparison but produces no handoff-related defects.

Suppliers	Inputs	Process	Outputs	Customers
Warehouse (inventory & flavors)	Customer order (verbally stated)	1. Customer joins queue and places order	Completed ice cream order	Walk-in customers
POS system	Ice cream product & containers	2. Customer decision time (flavor/item selection)	Accurate payment processed	General Manager
Scale / weighing equipment	Handwritten weight slip (busy period)	3. Maker scoops and weighs order	Improved queue throughput	Store Owner
Staff (Makers & Cashiers)	POS register	4. DEFECT RISK: Maker writes weight slip; passes to cashier		
		5. DEFECT RISK: Cashier reads slip, enters weight, processes payment		

4. The Measure Phase

4.1 Data Collection Plan — In-Person Observations

The team is currently conducting in-person observations at Beth Marie's on the Square during both peak and slow periods. The observation data collection plan tracks per-customer service time broken into component steps, item type, time of day, and staffing level. This granular breakdown enables waiting line model construction and bottleneck identification.

Metric	Unit	Collection Method	Purpose
Decision Time	Seconds	Stopwatch per customer	Measure time spent selecting flavor/item at counter
Scoop Time	Seconds	Stopwatch per customer	Measure order preparation and weighing time
Checkout Time	Seconds	Stopwatch per customer	Measure time from weight slip receipt to payment complete
Total Service Time	Seconds	Sum of above components	Overall throughput per customer for waiting line model
Item Selected	Category	Observer notation	Identify if item type affects service time (Cone, Cup, Split, Shake)
Time of Day	HH:MM	Logged per observation	Distinguish peak vs. slow period patterns
Number of Staff	Count	Logged per observation session	Correlate staffing level with service time and queue behavior

Note: Queue length at the time of customer arrival is not currently tracked but is recommended as an addition to future observation sessions. This metric is a key input for waiting line model accuracy, as it captures arrival rate relative to service rate — a core relationship in queuing theory.

What We Need to Measure

- Customer decision time — from joining the queue to decide on the items
- Scooping/prep time- How long it takes the employee to prepare the order

- Checkout time specifically — time spent at register from slip handoff to payment complete
- Service Rate — how long each step takes in both operating modes (This will be represented in Scooping time and Checkout time variables)
- Probability that no customers are in the system, P_0
- Average number of customers in the waiting line, L_q
- Average number of customers in the system, L
- Average time a customer spends in the waiting line, W_q
- Average time a customer spends in the system, W
- Probability an arriving customer has to wait, P_w

4.2 Data Collection Plan — Customer Perception Survey

In addition to in-person observational data, the team has developed a customer perception survey to capture subjective wait time experience and ordering behavior. The survey is currently pending approval from the store manager. Once approved, surveys will be administered on-site during observation sessions, enabling a direct comparison between measured service times and customer perception.

Survey Questions Asked

- Have you been to Beth Marie's before, or is this your first visit?
- How long do you feel your wait was today?
- Do you think your wait time was: Too long / Fair amount of time / Short?
- Did you pick a new flavor or one you have tried before?
- Do you have a go-to flavor at Beth Marie's?

Survey Purpose and Analysis Plan
The survey serves two analytical functions. First, it provides a customer satisfaction lens on the service time data — a wait that measures objectively short may still feel long to a customer who is undecided on flavor, and vice versa. Second, flavor selection behavior (new vs. familiar, go-to flavor) directly informs decision time variance, which is expected to be one of the higher-variability components in the waiting line model. Customers without a go-to flavor are hypothesized to exhibit significantly longer decision times, contributing disproportionately to queue length during peak periods.

4.3 Known Baseline Conditions

The following baseline conditions were established through the general manager interview and serve as the foundation for the measurement system design:

- Slow period: 1 staff member completes all steps — order, make, weigh, and ring up — with no handoff
- Busy period: Roles are split — dedicated cashier plus one or more makers
- Busy period handoff: Makers write weight on a paper slip and physically pass it to the cashier
- Known defect — misread weight: Incorrect number entered at register, resulting in wrong charge to customer
- Known defect — deciphering delay: Cashier pauses to read slip, slowing checkout and extending queue
- Known defect — slip loss/damage: Paper slip misplaced or illegible before reaching cashier; order must be re-weighed

- No documented standard exists for when to switch between the two operating modes
- Post-school and post-dinner rushes are the two predictable peak windows each day

4.4 Process Comparison — Where Value Is Lost

Step	Slow Period (Single Staff)	Busy Period (Split Role)
1. Order & Decision	Staff member takes order; customer decides at counter	Customer states order; decision time may back up queue
2. Scoop & Weight	Same staff member scoops and weighs	Maker scoops and weighs order
3. Handoff	No handoff — same person continues	DEFECT RISK: Maker writes weight on paper slip
4. Checkout	Same staff member rings up and charges	DEFECT RISK: Cashier reads slip, enters weight, processes payment
Defect Risk	Minimal — full ownership, no communication gap	High — legibility errors, deciphering delays, and slip loss all possible

4.5 Measurement System Analysis

Formal Gauge R&R analysis is not applicable at this stage as the primary measurements are observational time and event counts. The following considerations govern measurement system validity:

- Operational definitions: All metrics have written operational definitions before data collection. For example, decision time begins when the customer reaches the counter and ends when they verbally state their order.
- Observer consistency: Team members timing the same events will calibrate during an initial joint observation session to assess inter-rater reliability before independent data collection begins.

- POS data dependency: Billing error rate will ideally be drawn from POS records. If access is not approved, proxy measures using observation logs will be substituted.
- Observation windows: Data is being collected across multiple peak and slow periods to account for day-to-day variation in volume and staffing.
- Survey alignment: Customer survey responses will be cross-referenced with observed service times for the same time windows to assess perception accuracy.

4.6 Open Items and Data Gaps

Open Item	Impact	Resolution Path
POS data access — awaiting store owner approval	Limits ability to quantify historical billing error rate and transaction volume	Team to contact store owner; observation-based proxy measures identified as fallback
Customer survey — awaiting manager approval	Cannot collect perception data until permission is granted	Survey instrument is ready; team following up with manager
Queue length not currently tracked	Reduces precision of waiting line model arrival rate calculations	Recommended addition to observation sheet for future sessions
No documented mode-switching threshold	Cannot pre-determine when defect risk (split-role model) is active	Observe and document queue length or time-of-day trigger during site visits

5. AI Collaboration Documentation and AIAS Assessment

5.1 AI Use Overview

Artificial intelligence tools were used throughout the Define and Measure phases to support document synthesis, structured analysis, and report generation. All AI-generated content was reviewed, edited, and validated by the project team prior to inclusion in this report.

5.2 AI Interaction Log

Phase	AI Task	Human Input Provided	Human Validation Action
Define	Synthesize problem statement from raw interview notes and process analysis document	Uploaded two source documents; specified rubric requirements and formatting constraints	Team reviewed synthesized content against original documents for accuracy
Define	Generate Project Charter, SIPOC, VOC/CTQ table, and stakeholder analysis	Provided stakeholder list, process steps, known defect types, and team roles	Team verified stakeholder roles and CTQ specifications reflect operational reality
Measure	Structure observation data collection plan and customer survey section	Provided observation sheet column headers, survey questions, and tracking variables	Team reviewed sample design and operational definitions for feasibility
Report	Compile and format full report document per rubric specifications	Provided complete rubric, formatting requirements, all source content, and revision instructions	Team reviewed full document for completeness, accuracy, and rubric alignment

5.3 AIAS Level Assessment

Based on the AI Assistance Scale (AIAS), this project's use of AI assistance is assessed at Level 4. AI tools performed substantial synthesis, structure generation, and document creation tasks based on inputs provided by the team. However, all source data, problem framing, stakeholder identification, survey design, and validation decisions were made by human team members. The AI did not independently identify problems, collect data, or make recommendations without human direction.

AIAS Level	Justification
Level 4	AI performed substantial document synthesis and structured analysis tasks. The project team directed all tasks, provided all source information, and validated all AI outputs. AI served as a high-capability drafting and structuring tool, not as a decision-maker or independent analyst.

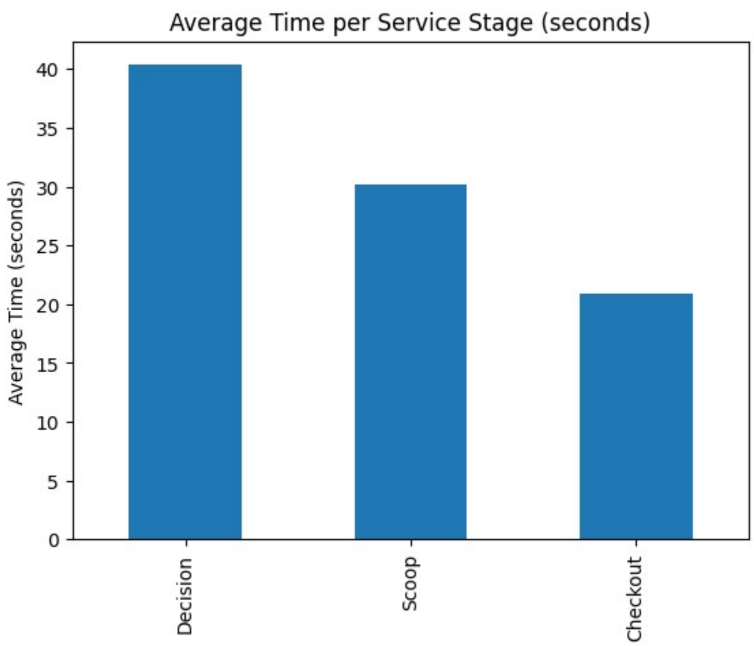
5.4 Human-AI Balance Reflection

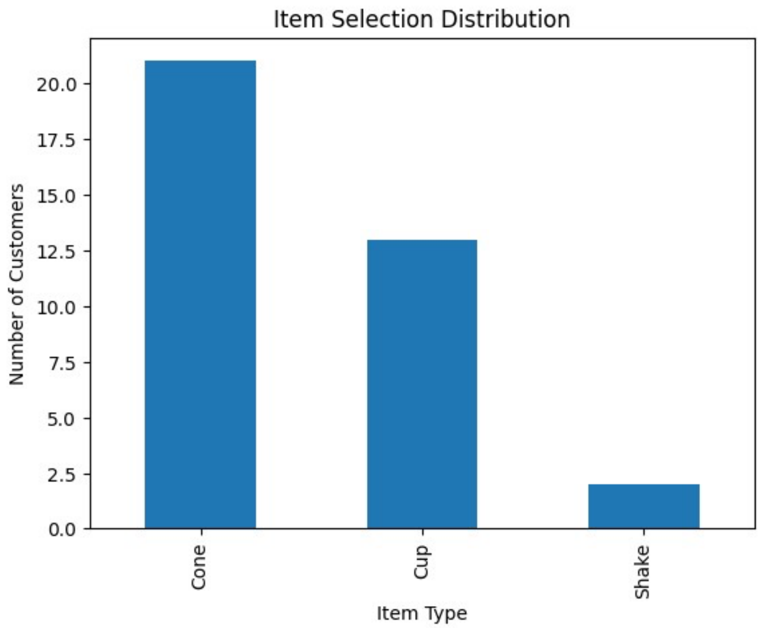
The team believes the balance of AI assistance in this project was appropriate to the task. Source data, observations, survey design, and problem identification were entirely human-generated through direct engagement with the general manager and operational context. AI was used to organize, synthesize, and format that information into a professional report structure. Future phases involving direct observation analysis, waiting line model construction, and improvement recommendations will similarly rely on human-led analysis with AI support for documentation and visualization.

Data Visualization and Analysis

The primary objective of this visualization was to observe the timing of the data that identifies bottlenecks, evaluate the variability of the items, and assess the work efficiency within Beth Marie's order process.

The service time was broken into Decision time, scoop time, and checkout time. These components were used to create bar charts, histograms, and trend analysis.



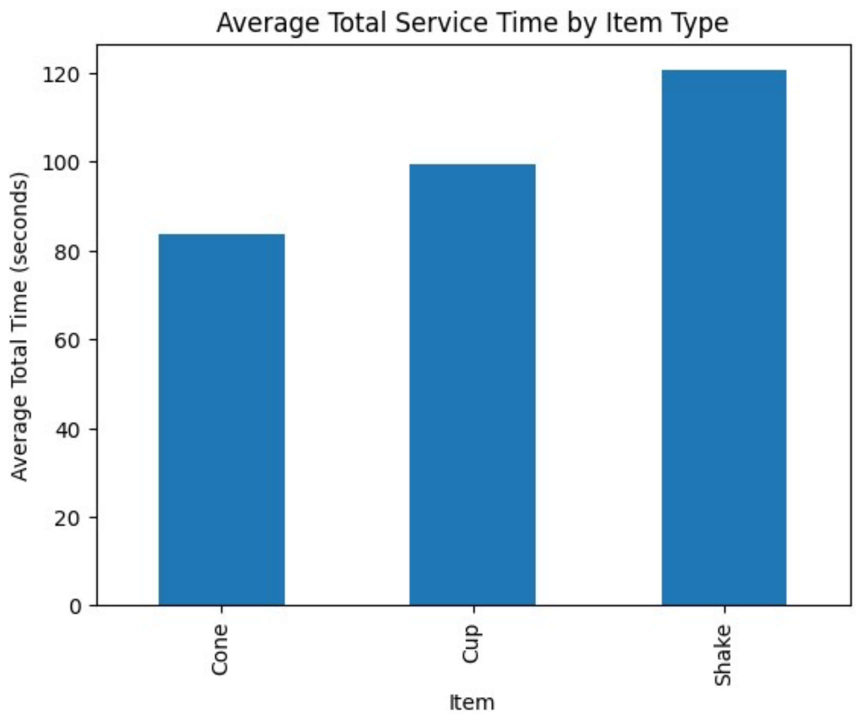


The bar chart displays that cones were ordered most frequently, and cups were the second most common. Even though Cones were ordered more frequently than Cups; Cups had a high coefficient of variance in both Decision Time (95.8%) and Scoop Times (70.6%) as demonstrated in the chart below.

Some customers ordered shakes as well, but they are very rare. Although Shakes represent only a small portion of the total orders, the service analysis bar chart shows that they required a longer preparation time. This also means that even though there was a small number of shake orders, they could reduce serving time during peak hours.

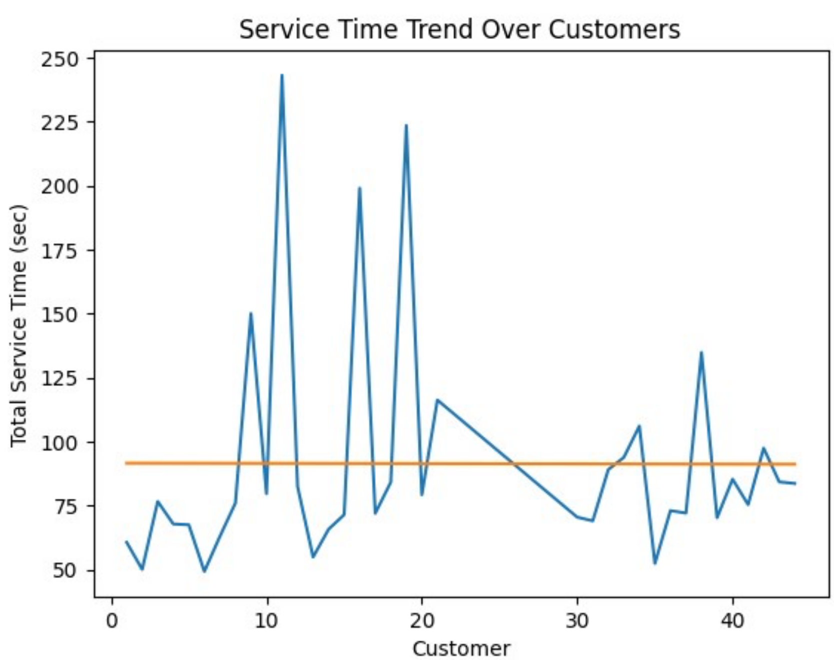
Average Time per Service Stage

This bar chart is a comparison of average time by service stage, showing that decision time is the longest stage, and the checkout time is the shortest, as well as consistent. Scoop time was moderate and depended on the number of servers.



Distribution of Total Service Time

A histogram of total service time shows that most customers fall within the 50–90 second range. Several outliers exceed 150 seconds. A small number exceeds 200 seconds.



Trend analysis was done by placing the total service time for each customer on a line graph and then adding a straight "trend" line to show the overall pattern. The graph shows how service time changes from one customer to the next. The trendline summarizes the general direction of the data.

In this case, the trendline is nearly flat, which means that there is no clear increase or decrease in service time over the observation period. Although there are several spikes where service time is much higher, these are isolated cases. Overall, the analysis shows that service times remain generally consistent, and the variation seen in the graph is caused by individual customers rather than a steady change in worker performance. From these visualizations, we can see that checkout time is efficient and stable, and the scoop time is moderate, but the decision time here plays a vital as well as a dominant role. This also shows that workers are not the root cause of the extended service time, but the customers' indecision and busy-period would delay the structure of the process.

The Improve Phase

Proposed Improvement Directions

Data collection procedure and waiting line model calculations will be used. The following section lists the possible areas of improvement based of the problem statement and our observations in measure phrase:

- We want to support the customers in picking flavors and make it easier for them.Since our data suggests that a major bottleneck is their long wait to select a flavors.Before they reach the front of the queue .We are currently researching approaches that will improve the general visibility of each flavor list.

- We aim to put together a basic Standard Operating Procedure (SOP) to increase the capacity of the rush hours. We are focusing on creating specific roles.This makes sure that the cashier has knowledge of exactly what needs to be double checked and is aware of where and when to pass the slip.

Implementation

Consumers often spend a long time choosing flavors which leads to a bottleneck during the ordering procedure and increases queues. Increasing general procedures effectiveness involves dealing with decision time, which has been determined to be the lengthiest and most unpredictable element in total service time.

To help customers make decisions more quickly, a kiosk device displays the entire list of available flavors and toppings, along with pictures and detailed descriptions.In displaying the popular flavors and seasonal options available could also help first-time visitors reducing hesitancy and deciding duration.
Adding a kiosk system can be completed in a short timeframe with proper planning. The process includes selecting a compatible kiosk provider, setting up the digital menu with flavors and toppings, and installing the device in-store. Once installed, the system is

tested and staff are briefly trained on usage and basic troubleshooting. Overall, implementation is simple and can be completed within a few weeks, helping improve efficiency and reduce customer wait times.

Control plan

The goal of the Control phase is to make sure that improvements continue to work over time and the process does not go back to being inefficient. The store will have to follow a clear and consistent way of handling orders during both slow and busy periods. A standard process should be created so employees know what to do in each situation. This process includes defining staff roles clearly during busy periods and setting a simple rule for when to switch from a single-staff system to a multi-staff system based on how busy the store is. This will help reduce confusion and keep the service running smoothly.

The store will also keep track of key performance measures such as total service time, decision time, checkout time, and customer wait time. These will be reviewed regularly to make sure the process is staying efficient. If wait times start to increase or errors occur more often, adjustments can be made quickly.

Employee training will be important to make sure everyone follows the new process correctly. Staff will be trained on their roles, the updated workflow, and any new systems, such as digital ordering or kiosks if they are implemented. This will help keep service consistent, especially during peak hours.

The store will also occasionally observe operations during busy periods to make sure the process is being followed. If any issues are noticed, they can be corrected right away.

Customer feedback will continue to be used to understand how people feel about their wait time and overall experience. This will help confirm that the improvements are working from both an operational and customer perspective.

Overall, this control plan ensures that the improvements are maintained by keeping the process consistent, monitoring performance, and making changes when needed. This will help Beth Marie’s continuing to reduce wait times and provide a better customer experience.

7. References

General Manager Interview Notes. (2024). Internal project document. Compiled by project team from in-person interview conducted at Beth Marie's on the Square, Denton, TX.

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American Society for Quality (ASQ). (2023). Six Sigma DMAIC methodology. Retrieved from https://asq.org/quality-resources/six-sigma

Pyzdek, T., & Keller, P. (2014). The Six Sigma Handbook (4th ed.). McGraw-Hill Education.

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The Appendices Portion

The following appendices contain supporting materials referenced in the main report.

Appendix content does not count toward the 5–10 page report limit.

Appendix A — Root Cause Analysis: Paper Slip Defect Types

- Defect Type 1 — Misread weight: Incorrect number entered at register, resulting in wrong charge to customer
- Defect Type 2 — Deciphering delay: Cashier pauses to read slip, slowing checkout and extending queue
- Defect Type 3 — Slip loss or damage: Paper slip misplaced or illegible before reaching cashier; order must be re-weighed

Appendix B — Observation Data Sheet Column Reference

Column	Unit	Definition
Customer	ID #	Sequential customer identifier per observation session
Decision Time	Seconds	Time from customer reaching counter to verbally stating order
Scoop Time	Seconds	Time from order stated to completed order placed on scale
Checkout Time	Seconds	Time from weight slip handoff (or weigh complete in slow period) to payment processed
Total Service Time	Seconds	Sum of Decision + Scoop + Checkout times
Item Selected	Category	Cone, Cup, Split, or Shake
Time of Day	HH:MM	Timestamp of observation
Number of Staff	Count	Total staff working on floor at time of observation

Appendix C — Proposed Improvement Directions (Pending Measure Phase Validation)

The following improvement concepts target root causes identified in the Define phase.

Final recommendations will be developed after full analysis of observation data.

- Replace handwritten slips with printed or digital weight display — eliminates legibility as a defect source entirely
- Standardize the mode-switching trigger — define a clear threshold (queue length, time of day, or customer count) for shifting from single-staff to split-role mode
- Create a standard operating procedure (SOP) for the busy-period handoff — even if paper slips remain short-term, a printed template with large pre-formatted fields reduces legibility errors
- Optimize role assignment during peak hours — define who owns which step to minimize idle time and queue backup
- Explore POS-integrated weighing — if the scale can connect to the POS system, weight is entered automatically, eliminating the handoff step entirely